# A CONGRUENCE SIEVE SURVIVOR SET WITHOUT LOGARITHMIC DENSITY (ERDŐS PROBLEM #486)

ARGUS OPEN RESEARCH PROJECT

Abstract. Erdős Problem #486 asks whether for every subset $A \subseteq \mathbb{N}$ and every family of residue classes $X_n \subseteq \mathbb{Z}/n\mathbb{Z}$ ($n \in A$), the survivor set $B = \{m \in \mathbb{N} : \forall\, n \in A \text{ with } n < m,\ m \bmod n \notin X_n\}$ possesses a well-defined logarithmic density. We document the formally verified counterexample constructing a fixed sparse sequence of moduli $A$ and residue sets $X_n$ whose survivor set $B$ has $\liminf_{x\to\infty} \frac{1}{\log x} \sum_{m\in B, m\le x} \frac{1}{m} \le \frac{177}{200} < \frac{49}{50} \le \limsup_{x\to\infty} \frac{1}{\log x} \sum_{m\in B, m\le x} \frac{1}{m}$.

## 1. Main Theorem

**Theorem 1.1** (Formally Verified in Lean 4)**.** *There exist a fixed infinite set $A \subseteq \mathbb{N}_{\ge 2}$ of rapidly growing coprime moduli and residue sets $X_n \subseteq \mathbb{Z}/n\mathbb{Z}$ such that the survivor set*

$$B \;=\; \left\{m \in \mathbb{N}_{\ge 1} : \forall\, n \in A,\ n < m \implies (m \bmod n) \notin X_n\right\}$$

*has logarithmic averages $L_B(x) = \frac{1}{\log x} \sum_{1\le m\le x,\, m\in B} \frac{1}{m}$ satisfying*

$$\liminf_{x\to\infty} L_B(x) \;\le\; \frac{177}{200} \;<\; \frac{49}{50} \;\le\; \limsup_{x\to\infty} L_B(x),$$

*and therefore $B$ does not possess a logarithmic density.*

## 2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 2141 in the domain of analytic and combinatorial number theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *formal artifact refutation* deciding the scope: fixed congruence sieve system with liminf L_B <= 177/200 < 49/50 <= limsup L_B (Lean 4 verified).

## 3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- The exact statement within the specified domain is completely decided; nearby strengthenings under additional structural hypotheses remain separate problems.

## References

[1] Primary problem list and source literature for Result 2141 (analytic and combinatorial number theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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